

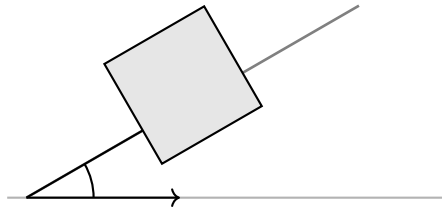
Physics Packet: Friction on an Inclined Plane

Name: _____

Date: _____

Part A: Review of Forces on an Inclined Plane

A box is on an incline making an angle θ with the horizontal.

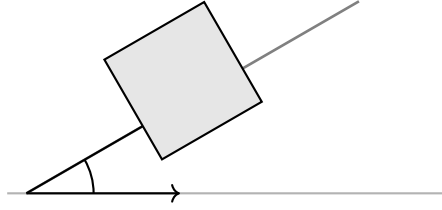


1. How does the Free Body Diagram look like?
2. What are the force summations?
3. Which force is responsible for causing motion *along* the incline?
4. Which force prevents the box from falling *through* the surface?
5. Is the normal force equal to the weight of the box on an incline? Why or why not?

Part B: Static Friction on an Inclined Plane (No Motion)

A box is on an incline making an angle θ with the horizontal.

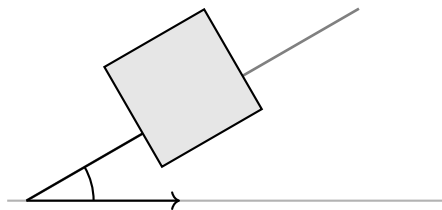
1. How does the Free Body Diagram look like?



2. When the box is at rest, what must be true about the net force along the incline?
3. Does static friction always equal $\mu_s F_N$? Explain.
4. What determines the *actual* value of the static friction force?
5. As the angle of the incline increases, what happens to the required static friction force?
6. What are the force summations?

Part C: Kinetic Friction on an Inclined Plane (Motion)

A box is on an incline making an angle θ with the horizontal.



1. How does the Free Body Diagram look like?
2. What condition must be met for the box to begin sliding?
3. At the instant the box starts to move, how does the friction force change?
4. What are the force summations?

Practice Problems

Problem 1

A 5 kg box rests on a 25° incline. $\mu_s = 0.60$.

1. Draw a Free Body Diagram. (FBD)
2. Find F_N , F_g , F_{g_x} and F_{g_y} .
3. Find the maximum static friction force F_{f_s} .
4. Does the box slide? Explain.
5. If it moves, find the kinetic friction force.
6. Find $\sum F_x$ and $\sum F_y$.
7. Find the acceleration of the box.

Problem 2

A 10 kg box rests on a 30° incline. $\mu_s = 0.35$, $\mu_k = 0.25$.

1. Draw a Free Body Diagram. (FBD)
2. Find F_N , F_g , F_{gx} and F_{gy} .
3. Find the maximum static friction force F_{fs} .
4. Does the box slide? Explain.
5. If it moves, find the kinetic friction force.
6. Find $\sum F_x$ and $\sum F_y$.
7. Find the acceleration of the box.

Problem 3

A 13 kg box rests on a 33° incline. $\mu_s = 0.55$.

1. Draw a Free Body Diagram. (FBD)
2. Find F_N, F_g, F_{gx} and F_{gy} .
3. Find the maximum static friction force F_{fs} .
4. Does the box slide? Explain.
5. If it moves, find the kinetic friction force.
6. Find $\sum F_x$ and $\sum F_y$.
7. Find the acceleration of the box.

Problem 4

A 122 kg box rests on a 60° incline. $\mu_s = 0.85$, $\mu_k = 0.70$.

1. Draw a Free Body Diagram. (FBD)
2. Find F_N , F_g , F_{gx} and F_{gy} .
3. Find the maximum static friction force F_{fs} .
4. Does the box slide? Explain.
5. If it moves, find the kinetic friction force.
6. Find $\sum F_x$ and $\sum F_y$.
7. Find the acceleration of the box.